

Def.

Given four complex numbers $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$, a map

$$w = T(z) = \frac{az + b}{cz + d}$$

is called a linear fractional transformation.

Observation for the linear fraction.

The condition $ad - bc \neq 0$ gives that the $T(z)$ is non-constant, assume $ad - bc = 0$.

If $c \neq 0$, then

$$\frac{az + b}{cz + d} = \frac{\frac{a}{c} \cdot (cz + d) + b - \frac{ad}{c}}{cz + d} = \frac{a}{c}.$$

If $a \neq 0$, then

$$\frac{az + b}{cz + d} = \frac{az + b}{cz + \frac{bc}{a}} = \frac{a}{c} \cdot \left(\frac{z + \frac{b}{a}}{z + \frac{bc}{a^2}} \right) = \frac{a}{c} \quad \left(z \neq -\frac{b}{a} \right).$$

If $a = 0$, then either $b = 0$ or $c = 0$.

If $b = 0$, then $T \equiv 0$.

If $c = 0$, then $T \equiv \frac{b}{d}$.

If $c = 0$, then either $a = 0$ or $d = 0$.

If $a = 0$, then $T \equiv \frac{b}{d}$.

If $d = 0$, then T is not defined.

Now, we assume $ad - bc \neq 0$, then on $\mathbb{C} \setminus \{-\frac{d}{c}\}$.

$$w = T(z) = \begin{cases} \frac{a}{c} - \left[\frac{ad - bc}{c^2} \right] \cdot \frac{1}{z + \frac{d}{c}} & \text{if } c \neq 0 \\ \frac{a}{d}z + \frac{b}{d} & \text{if } c = 0, \end{cases}$$

Therefore, the T is injective.

